PURITY ALONG A GEOMETRIC POINT: FOUR EQUIVALENT FORMULATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a limit argument descending purity from the etale local ring to an etale neighbourhood, and an ascent along a locally quasi-finite base change using that a quasi-finite morphism over a strictly henselian base becomes finite

Editorial note. The label lemma-pure-along-X-y names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Impurities as witnesses that a module is not pure along a point, and the definition of purity along a point in terms of the associated points of the geometric fibres
- The etale local ring at a geometric point as the filtered colimit of the rings of etale neighbourhoods, and the limit characterisation of purity that follows from it
- Ascent of an impurity along a locally quasi-finite base change, and the fact that a quasi-finite morphism over a strictly henselian base becomes finite after passing to an open and closed neighbourhood
- Base change of the relative assassin under a finite morphism, and closedness of the induced map on underlying spaces

Source reference. Original reference 0CVA: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 366–366. Prerequisite pointer invoked at proof line(s) 444; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05IZ: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 319–319. Prerequisite pointer invoked at proof line(s) 433; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02LK: sources/stacks-chow/tags.txt, source lines 3411–3411. Prerequisite pointer invoked at proof line(s) 454; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-flat.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01TL: sources/stacks-chow/tags.txt, source lines 2378–2378. Prerequisite pointer invoked at proof line(s) 451; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05FW: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 260–260. Prerequisite pointer invoked at proof line(s) 403; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CV2: sources/stacks-chow/tags.txt, source lines 16122–16122. Prerequisite pointer invoked at proof line(s) 460; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03XL: sources/stacks-chow/tags.txt, source lines 5152–5152. Prerequisite pointer invoked at proof line(s) 450; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03XU: sources/stacks-chow/tags.txt, source lines 5161–5161. Prerequisite pointer invoked at proof line(s) 438; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04JW: sources/stacks-chow/tags.txt, source lines 5952–5952. Prerequisite pointer invoked at proof line(s) 430; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 05FW. Let $\overline{y}$ be a geometric point lying over y. Let $\mathcal{O} = \mathcal{O}_{Y,\overline{y}}$ be the étale local ring of Y at $\overline{y}$. Denote $Y^{sh} = \mathrm{Spec}(\mathcal{O})$, $X^{sh} = X \times_Y Y^{sh}$, and $\mathcal{F}^{sh}$ the pullback of $\mathcal{F}$ to X^{sh}. The following are equivalent*

1. *there exists an impurity $(Y^{sh} \to Y, y' \rightsquigarrow \overline{y}, \xi)$ of $\mathcal{F}$ above y,*
2. *every point of $Ass_{X^{sh}/Y^{sh}}(\mathcal{F}^{sh})$ specializes to a point of the closed fibre $X_{\overline{y}}$,*
3. *there exists an impurity $(T \to Y, t' \rightsquigarrow t, \xi)$ of $\mathcal{F}$ above y such that $(T, t) \to (Y, y)$ is an étale neighbourhood, and*
4. *there exists an impurity $(T \to Y, t' \rightsquigarrow t, \xi)$ of $\mathcal{F}$ above y such that $T \to Y$ is quasi-finite at t.*

Proof. That parts (1) and (2) are equivalent is immediate from the definition.

Recall that $\mathcal{O} = \mathcal{O}_{Y,\overline{y}}$ is the filtered colimit of $\mathcal{O}(V)$ over the category of étale neighbourhoods $(V, \overline{v}) \to (Y, \overline{y})$ (Properties of Spaces, Lemma 04JW). Moreover, it suffices to consider affine étale neighbourhoods V. Hence $Y^{sh} = \mathrm{Spec}(\mathcal{O}) = \lim \mathrm{Spec}(\mathcal{O}(V)) = \lim V$. Thus we see that (1) implies (3) by Lemma 05IZ.

Since an étale morphism is locally quasi-finite (Morphisms of Spaces, Lemma 03XU) we see that (3) implies (4).

Finally, assume (4). After replacing T by an open neighbourhood of t we may assume $T \to Y$ is locally quasi-finite. By Lemma 0CVA we find an impurity $(T_1 \to Y^{sh}, t_1' \rightsquigarrow t_1, \xi_1)$ with $T_1 \to T \times_Y Y^{sh}$ étale. Since an étale morphism is locally quasi-finite and using Morphisms of Spaces, Lemma 03XL and Morphisms, Lemma 01TL we see that $T_1 \to Y^{sh}$ is locally quasi-finite. As $\mathcal{O}$ is strictly henselian, we can apply More on Morphisms, Lemma 02LK to see that after replacing T_1 by an open and closed neighbourhood of t_1 we may assume that $T_1 \to Y^{sh} = \mathrm{Spec}(\mathcal{O})$

is finite. Let $\theta \in |X^{sh}|$ be the image of ξ_1 and let $y' \in \mathrm{Spec}(\mathcal{O})$ be the image of t_1'. By Divisors on Spaces, Lemma 0CV2 we see that $\theta \in \mathrm{Ass}_{X^{sh}/Y^{sh}}(\mathcal{F}^{sh})$. Since $\pi : X_{T_1} \to X^{sh}$ is finite, it induces a closed map $|X_{T_1}| \to |X^{sh}|$. Hence the image of $\overline{\{\xi_1\}}$ is $\overline{\{\theta\}}$. It follows that $(Y^{sh} \to Y, y' \rightsquigarrow \overline{y}, \theta)$ is an impurity of $\mathcal{F}$ above y and the proof is complete. □